



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

DATA ENCRYPTION AND FILE SECURITY TOOL

The domain of the Project:
CYBER SECURITY

Team Mentors (and their designation):
HARI HARAN (SECURITY OPERATOR)

Team Members:
MS. SNEHA ROY

Period of the project
6 MONTHS

November 2025 to May 2026



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Declaration

The project titled “Data Encryption and File Security Tool” has been mentored by Mr HariHaran Sir, organised by SURE Trust, from November 2025 to May 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members: Sneha Roy

Ms. ----- -- Signature

Mentor’s Name Mr HariHaran
Designation—Company Name

Mentor’s Name
Designation—Company Name

Prof. Radhakumari
Executive Director & Founder
SURE Trust



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Table of contents

1. Executive summary
2. Introduction
3. Project Objectives
4. Methodology & Results
5. Social / Industry relevance of the project
6. Learning & Reflection
7. Future Scope & Conclusion



Executive Summary

- This project, Advanced Data Encryption and File Security Tool, focuses on securing digital files using modern cryptographic techniques such as AES-256 encryption, RSA encryption, and hybrid encryption. The objective is to provide a reliable and user-friendly system that ensures data confidentiality and protection against unauthorized access. The system allows users to encrypt and decrypt files using different methods, including symmetric encryption, password-based encryption, and a hybrid approach combining AES and RSA for enhanced security. The methodology involves secure key generation, file processing, and a graphical user interface for easy interaction.
- The project highlights the implementation of strong encryption standards, efficient file handling, and multiple security layers to ensure both performance and protection. It demonstrates how combining fast encryption (AES) with secure key exchange (RSA) can create a robust system suitable for real-world applications. The tool is designed to be simple yet powerful, making it useful for individuals and organizations that require secure data storage and transfer. It also emphasizes practical learning in cybersecurity and can be further extended for advanced applications such as cloud security and secure communication systems.



Introduction

- **Background and context of the project:** - In today's digital era, the security of data has become a critical concern due to the increasing risk of cyber threats, unauthorized access, and data breaches. With the growing use of digital files for personal, academic, and professional purposes, it is essential to ensure that sensitive information remains protected. This project focuses on developing a secure file encryption tool that helps users safeguard their data using modern cryptographic techniques
- **Problem statement or goals of the project:** - The main problem addressed in this project is the lack of simple and accessible tools for securely encrypting and decrypting files. Many users do not have the technical knowledge to implement advanced security measures, which makes their data vulnerable. The goal of this project is to provide a user-friendly application that enables secure file encryption using AES-256, RSA, and hybrid encryption methods, ensuring both efficiency and strong protection.
- **Scope and limitations of the project:** - The scope of this project includes file-based encryption and decryption using different methods through a graphical user interface. It allows users to select files, apply encryption, and restore them when needed. However, the project is limited to local file encryption and does not include features such as cloud integration, network-based security, or multi-user access control.
- **Innovation component in the project:** - The innovation in this project lies in the integration of multiple encryption techniques within a single system, particularly the use of hybrid encryption, which combines the speed of AES with the security of RSA. This approach enhances both performance and data protection, making the system more robust and practical for real-world applications.



Project Objectives

- The primary objective of this project is to develop a secure and efficient file encryption tool that protects sensitive data from unauthorized access. The system aims to implement strong cryptographic techniques such as AES-256, RSA, and hybrid encryption to ensure both speed and high security. Another key objective is to design a user-friendly graphical interface that allows users to easily select files, apply encryption or decryption, and manage their data without requiring advanced technical knowledge.
- The expected outcome of this project is a fully functional application capable of encrypting and decrypting files using multiple methods, including symmetric, password-based, and hybrid encryption. The deliverables include a working software tool, properly encrypted output files, decrypted original files, and a reliable system that ensures data confidentiality and integrity. The project also aims to demonstrate practical implementation of cybersecurity concepts that can be applied in real-world scenarios.



Methodology and Results

- **Methods/Technology used:**

The project uses modern cryptographic techniques including AES-256 (CBC mode) for fast symmetric encryption, RSA (2048-bit) for secure key encryption, and hybrid encryption which combines both methods. Password-based encryption is implemented using PBKDF2 for generating secure keys from user passwords.

- **Tools/Software used:**

The system is developed using Python programming language. The graphical user interface is built using Tkinter, and the encryption functionalities are implemented using the cryptography library. The project is developed and tested using Visual Studio Code.

- **Project Architecture (Clearly describe the architecture)**

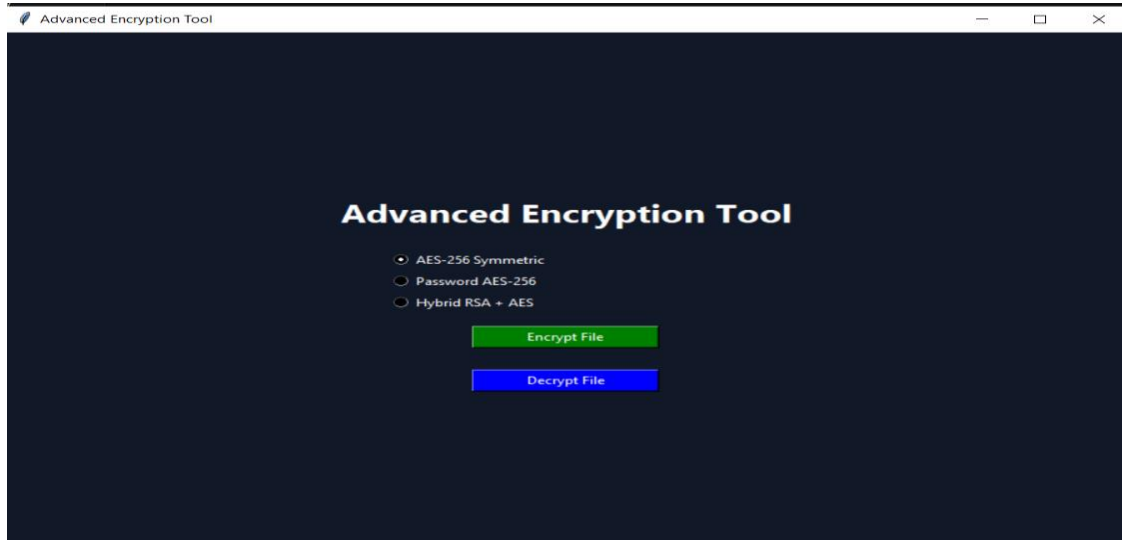
The system follows a layered architecture:

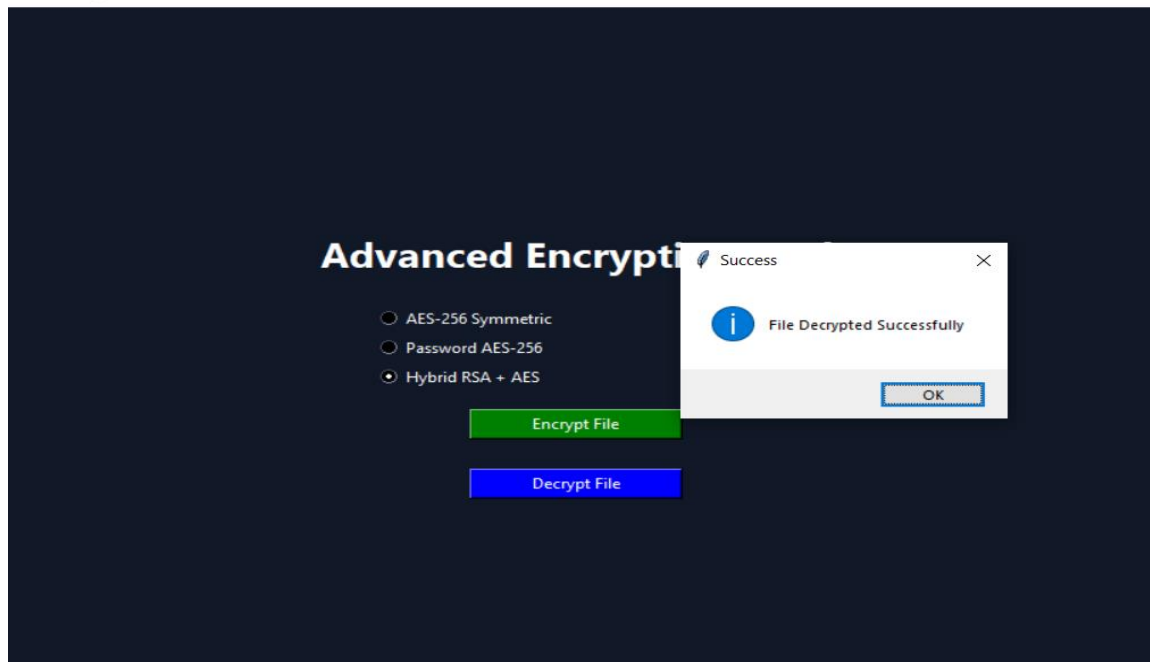
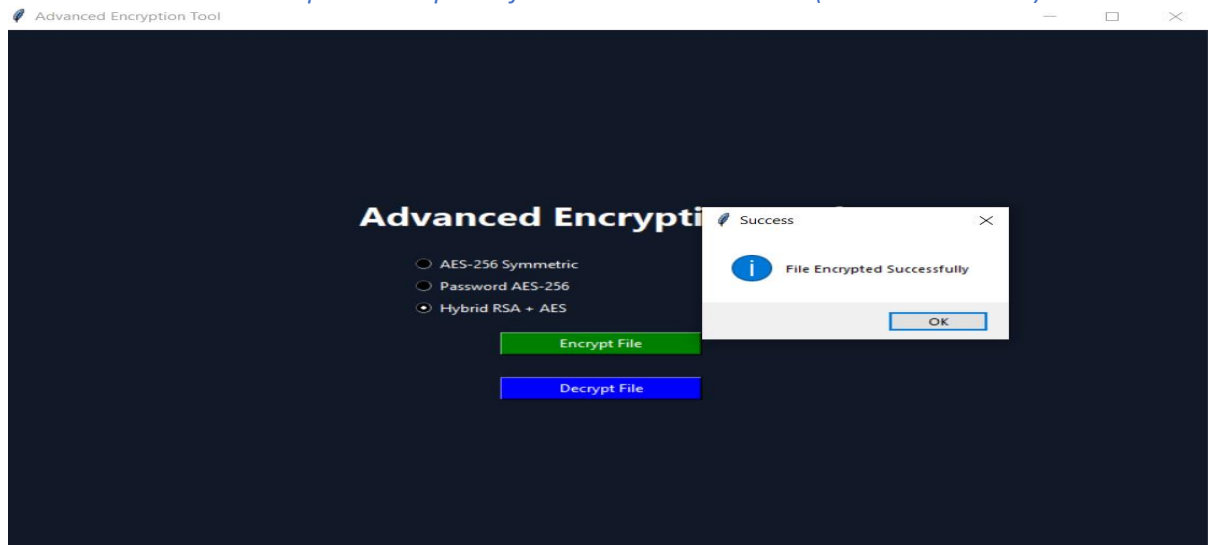
User Interface Layer: Built using Tkinter for user interaction

Application Logic Layer: Handles encryption, decryption, and file operations

Encryption Layer: Implements AES, RSA, and hybrid encryption algorithms

- **Final project working screenshots along with supporting explanation**





The application allows users to select a file and choose an encryption method (AES, password-based, or hybrid). Upon clicking the encrypt button, the file is securely encrypted and saved. During decryption, the user selects the encrypted file and provides the required key or password to restore the original file. (Insert screenshots here showing UI, encryption, and decryption process.)

- Project GitHub Link <https://github.com/sure-trust/SNEHA-ROY-g14-cs>



- **Learning (Technology & Skills)**

During the development of this project, I gained practical knowledge of cryptographic techniques such as AES-256 encryption, RSA encryption, and hybrid encryption. I also learned how to implement secure key management, password-based encryption using PBKDF2, and file handling in Python. Additionally, I improved my skills in GUI development using Tkinter and gained experience in debugging, error handling, and structuring a complete application.

- **Reflection (Overall Experience)**

Working individually on this project helped me develop problem-solving skills, time management, and independent decision-making. I faced challenges related to encryption logic, key handling, and debugging errors, which improved my understanding of real-world security implementations. Overall, this project was a valuable learning experience that strengthened both my technical knowledge and confidence in developing secure software applications.



Conclusion and Future Scope

- **Conclusion**

This project successfully achieved its objective of developing a secure file encryption tool using AES-256, RSA, and hybrid encryption techniques. The system provides a user-friendly interface for encrypting and decrypting files efficiently while ensuring data confidentiality. The implementation demonstrates how combining fast encryption (AES) with secure key protection (RSA) results in a robust and reliable security solution. Overall, the project highlights the practical application of cryptography in protecting digital data.

- **Future Scope**

1. Integration with cloud storage for secure online file protection
2. Development of a mobile or web-based version of the application
3. Implementation of advanced encryption modes like GCM for enhanced security
4. Addition of multi-user authentication and access control
5. Real-time secure file sharing and communication features